Universitat de Barcelona

Master in Data Science

Time Series Analysis

Quantitative Assessment of the Russian Invasion of Ukraine Using a Counterfactual Analysis

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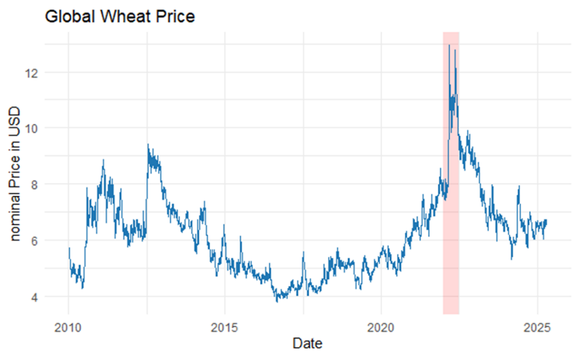
Contents

1. Introduction

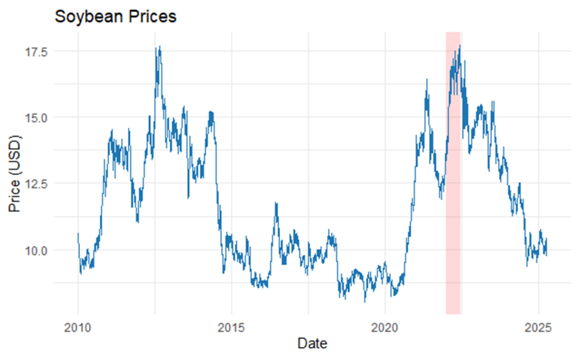
1.1 Context

The global economy is intricately linked to the stability of commodity markets, with agricultural staples such as wheat playing a pivotal role in daily life. As a key ingredient in both basic and processed foods, fluctuations in wheat prices have far-reaching consequences, extending beyond the dinner table to impact services such as transportation and distribution, thereby creating ripple effects across the entire economic landscape. Following the onset of the war in Ukraine, a significant and sustained increase in global commodity prices, including wheat, has been observed. This project aims to quantitatively assess whether this widespread price inflation can be directly attributed to the conflict.

This study seeks to predict the effect of the Russian invasion of Ukraine on global wheat prices. We acknowledge that since the war began, all prices have increased significantly, and a central objective is to analyze whether this phenomenon is a direct consequence of the conflict or attributable to other confounding factors. To achieve this, our analysis will examine how various factors, including wheat production levels, prevailing weather conditions, the total cultivated land area, and underlying price dynamics, collectively impact global food inflation. Specifically, this research will measure the impact of the 2022 Russian invasion of Ukraine on wheat prices and production within Europe. Utilizing historical data spanning from 2000 to 2024, we will construct an optimal linear model based on carefully selected variables. This model will serve as a baseline to predict wheat market behavior prior to the invasion. By comparing the model's predictions for February 2022 (the month of the invasion) with the actual observed market values, we can precisely identify any significant deviations. This counterfactual analysis will allow us to attribute these deviations to the war's impact, thereby quantifying its economic consequences on the global wheat market.

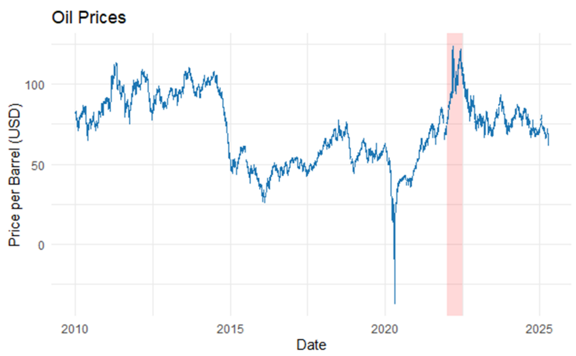


(a) Wheat prices



(b) Corn prices

(c) Soybean prices



(d) Oil prices

Figure 1: Commodity price trends (2010-2024)

1.2 Causal Inference

Understanding the true impact of a singular event, such as the Russian invasion of Ukraine, on complex global systems like commodity markets presents a significant challenge. While observed price increases following the conflict are evident, establishing a definitive causal link requires rigorous analytical methods that move beyond mere correlation. The primary goal of this study is to isolate and quantify the causal effect of the war on global wheat prices, disentangling it from other contemporaneous factors that might also influence market dynamics.

Traditional econometric models often describe associations, but to determine whether the war caused the observed changes, we must employ techniques from the field of causal inference (Shelby Temple, 2020 and Zhiying Gu, 2020). Our approach hinges on the concept of a counterfactual: **what would the global wheat price have been in the absence of the Russian invasion, all else being equal?** Since directly observing this counterfactual scenario is impossible, we must construct a plausible estimate. This paper will leverage counterfactual analysis to simulate a "no-war" scenario by building a predictive model of wheat price behavior based on pre-invasion trends and relevant covariates. By comparing the actual observed wheat prices after February 2022 with the prices predicted by our counterfactual model, we can estimate the specific causal impact attributable to the conflict. This methodology allows us to move beyond simple before-and-after comparisons, providing a more robust and statistically sound assessment of the war's economic consequences on the wheat market.

1.3 Counterfactual Analysis or Basket of Controls Modeling

Following the principles outlined in the previous section on causal inference, a critical challenge in establishing a robust counterfactual lies in the careful selection of control variables. Our aim is to construct a model whose explanatory variables are not themselves significantly influenced by the war, ensuring that the model truly represents a "no-war" scenario.

If we were to include variables such as global corn prices or soybean prices in our predictive model (as illustrated in Figure 1b and Figure 1c respectively), we face a significant methodological pitfall. Given the systemic nature of the war's impact on global commodity markets, it is highly probable that corn and soybean prices also reflect, to some degree, the war's influence. Incorporating such variables would inadvertently "mimic" the war's impact within our baseline model. This means that even without explicitly including a war "dummy variable" or a direct measure of the conflict, the model would inherently incorporate the war's effects through these contaminated control variables. Consequently, the comparison between our model's predictions and actual observed wheat prices would yield a biased estimate of the war's specific impact, as the counterfactual itself would already embody some of the effect we are trying to isolate. Therefore, a meticulous selection of control variables that are demonstrably exogenous to the war's direct commodity market effects is paramount for constructing a valid counterfactual.

To visualize this challenge, Figure 2 presents two linear models. One model includes all available variables, along with a specific dummy variable for the war. The other model, however, does not explicitly include this war dummy. The selection of regressors for both models is automated via stepwise regression based on the Akaike Information Criterion (AIC). As shown in Figure 2, both models exhibit a strikingly similar pattern. This shows that regardless of whether the war is explicitly declared as a regressor, its pervasive influence is already captured by other variables that have themselves been impacted by the conflict. Therefore, a meticulous selection of control variables that are demonstrably exogenous to the war's direct commodity market effects is paramount for constructing a valid counterfactual.

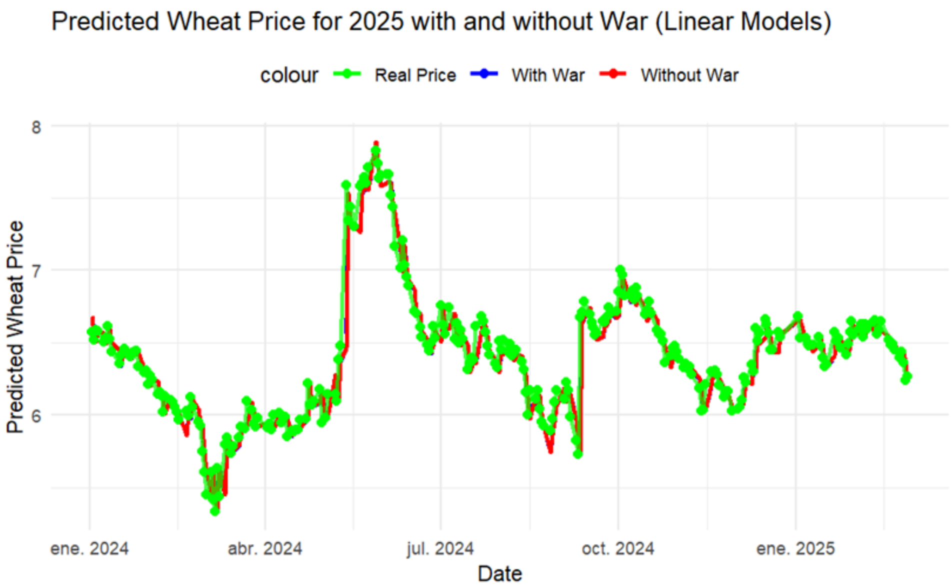


Figure 2: Predicted Wheat Prices with all available variables with war and without war dummy variable

1.4 Data

The data used for this report is based on different daily datasets, including commodity prices, weather variables, and economic indicators. All time series cover the period 2010-2025 unless otherwise specified, with sources from Macrotrends, ECA&D, and official agencies. To asses the analysis, the period used has been from 2010 to 2024.

Series	**Period**	**Region**	**Unit**	**Source**	**Notes**
Wheat Price	2010-2025	Global	USD/Bushel	macrotrends.net/2534	Monthly averages
Corn Price	2010-2025	Global	USD/Bushel	macrotrends.net/2532	US No. 2 Yellow Corn
Soybean Price	2010-2025	Global	USD/Bushel	macrotrends.net/2531	-
Precipitation	2010-2025	Ukraine	0.1 mm	ecad.eu	3 major cities
Mean Temp	2010-2025	Ukraine	°C	ecad.eu	Original in 0.1°C
Max Temp	2010-2025	Ukraine	°C	ecad.eu	Original in 0.1°C
Natural Gas	2010-2025	Global	USD/MMBtu	macrotrends.net	-
Crude Oil	2010-2025	Global	USD/Barrel	eia.gov	-
Baltic Dry	2010-2025	Global	Index	investing.com	Shipping costs Index
EUR/USD	2010-2025	Global	EUR/USD	fred.stlouisfed.org	Daily averages
Crop Stats	2010-2024	Global/Ukraine	HA/MT	fas.usda.gov	Area/Yield/Prod
Copper Price	2010-2025	Global	USD/lb	macrotrends.net	LME prices
Sugar Price	2010-2025	Global	USD/lb	macrotrends.net	ICE No. 11
Oats Price	2010-2025	Global	USD/Bushel	macrotrends.net	-
Coffee Price	2010-2025	Global	USD/lb	macrotrends.net	Arabica futures
Lumber Price	2010-2025	Global	USD/1000bdft	macrotrends.net	Random lengths

Table 1: Data used

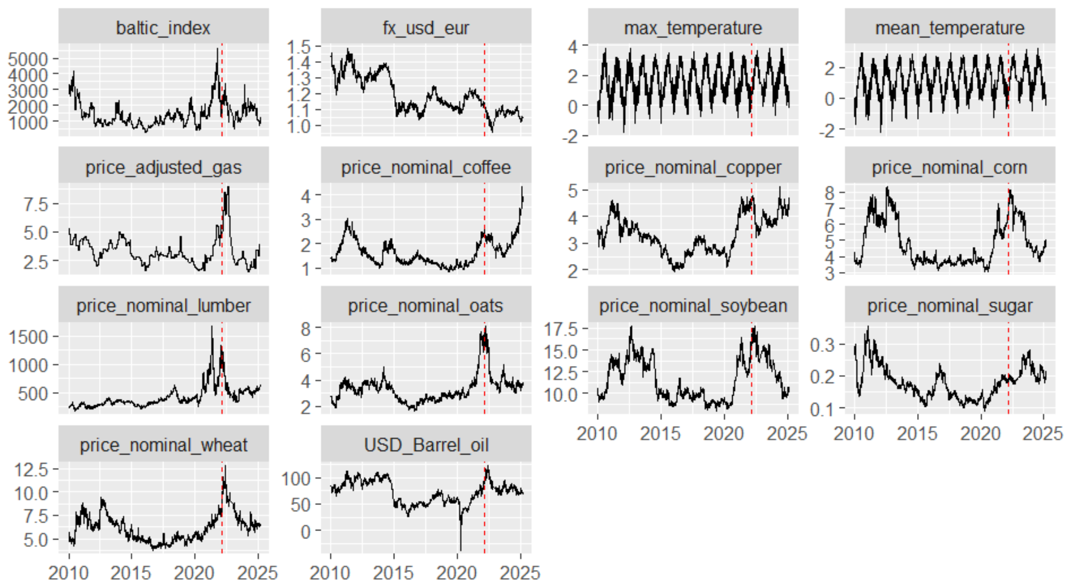


Figure 3: Example of some time series used

Our dataset, that contains historical information from 2000 to 2024, has been submitted to diverse pre-processing steps to ensure data quality.

- Geographic Data Consolidation: To provide a comprehensive view of relevant agricultural regions, precipitation data was specifically sourced for Kyiv. Additionally, to fill missing values and to account for broader regional influences, data from Mykolaiv (NAS) was strategically merged with that of Odesa, and similarly, Mykolaiv data was combined with Ternopil. This aggregation helps capture localized weather patterns and their potential impact on agricultural output.
- Handling Missing Values: For isolated missing values on specific dates, a dedicated imputation function was applied. This function effectively addresses gaps by populating missing entries with the corresponding value from the previous year on the same day. This method is particularly useful for cyclical data like weather patterns, where year-over-year consistency is often observed.
- Annual Data Transformation: For variables where data was available only on an annual basis rather than daily, the annual value was uniformly applied to each day within that respective year. This approach allows for the integration of annual trends into our daily-level analysis.
- Price Normalization: All price data is presented in nominal USD. This ensures that the prices are directly comparable across the entire study period, allowing for an accurate assessment of price fluctuations in relation to inflation.
- Temperature Unit Conversion: Finally, all temperature measurements were converted to degrees Celsius to maintain consistency and facilitate straightforward interpretation within the model.

2. Methodology

2.1 Quantifying the impact

Let y denote the target variable, representing the global wheat price in dollars. Define the temporal domain as $\mathbb{T} = \mathbb{T}_0 \cup \mathbb{T}_1$, where $\mathbb{T}_1$ spans from February 24, 2022 — the date of the invasion — to February 24, 2023, and $\mathbb{T}_0$ refers to the historical period preceding the invasion.

Let $X = (x_1, \dots, x_n) = X = \{(x_{1,t}, \dots, x_{n,t})\}_t$ denote the multivariate time series of explanatory variables. Consider a model $f(x_1, \dots, x_n) = \hat{y}$, trained on data from $\mathbb{T}_0$, to forecast the target variable y. The prediction error is defined as $\varepsilon = \hat{y} - y$.

Suppose that, for all $t \in \mathbb{T}_0$, the model residuals follow a normal distribution, i.e., $\varepsilon \sim \mathcal{N}(0, \sigma^2)$. Let $p \in (0, 1)$, and choose ε_0 such that

$$P\left(\left|\mathcal{N}(0, \sigma^2)\right| < \varepsilon_0\right) > p.$$

Then, a confidence interval can be constructed for the forecasted series: with probability at least p, the true value y will lie within the interval $[\hat{y} - \varepsilon_0, \hat{y} + \varepsilon_0]$.

For the post-invasion period $\mathbb{T}_1$, we model the target variable as

$$y_t = \hat{y}_t + \varepsilon + W,$$

where W is a real-valued random variable representing the impact of the war. We make no assumptions about the distribution of W, nor do we intend to model it explicitly. Naturally, in the absence of war, we assume $W = 0$, so the expression $y = \hat{y} + \varepsilon + W$ remains valid throughout the entire timeline. Our key assumption is that the distribution of ε remains the same in both $\mathbb{T}_0$ and $\mathbb{T}_1$, and that ε and W are independent.

Since both ε and W are random variables and are summed together, it is impossible to observe them separately. However, because the distribution of ε is known, we can bound it with high confidence. Specifically, for a chosen confidence level p, we know that $|\varepsilon| < \varepsilon_0$ with probability p. This allows us to construct a confidence bound for W as well. Explicitly:

$$\left.\begin{array}{l} W = y - \hat{y} - \varepsilon \\ -\varepsilon_0 < \varepsilon < \varepsilon_0 \end{array}\right\} \implies \begin{array}{l} y - \hat{y} - \varepsilon_0 < W < y - \hat{y} + \varepsilon_0 \\ \text{with confidence level } p. \end{array}$$

Therefore, knowing the series $\{y_t\}$ and $\{\hat{y}_t\}$ for $t \in \mathbb{T}_1$, along with ε_0, we can compute this confidence interval for W. We will try to be as conservative as possible, so we will always assign the value to W that is the furthest away from our hypothesis at confidence p. In this way, we will be able to say that *the war had an impact of at least this value*. With this being said, from this confidence interval, we will select the value for W that is the lowest in absolute value. So, if the model failed within $[-\varepsilon_0, \varepsilon_0]$, we will consider that the war had no impact and $W = 0$. If our model underestimated the price by more than ε_0, we will assign $W = y - \hat{y} - \varepsilon_0$ and say that the war impact on that period was higher than W at a confidence of p. And conversely for when our model overestimates the price.

Finally, the estimated average impact of the war during the first year will be computed differently depending on whether we are working with the original wheat prices series or its first differences.

In case it is the original price series, the impact will be given by:

$$\frac{1}{|\mathbb{T}_1|}\sum_{t\in\mathbb{T}_1} W_t$$

which can be understood as: *the average of the abnormal gap between the real series and the predicted series.*

Otherwise, if we are working with its first differences, it will be:

$$\sum_{t\in\mathbb{T}_1} W_t$$

which can be understood as: *the war yielded wheat price increases that together account for at least this value.*

2.2 Assumptions

We state our assumptions as follows:

1. At time $\mathbb{T}_0$, the error $\varepsilon = y - \hat{y}$ is normally distributed:

   $$\varepsilon \sim \mathcal{N}(0, \sigma^2).$$

   The value of σ will be determined based on the observed data.

2. At time $\mathbb{T}_1$, the error ε also follows:

   $$\varepsilon \sim \mathcal{N}(0, \sigma^2).$$

3. The error ε and the variable W are independent.

2.3 Building the model

2.3.1 Interpolating missing values and correcting frequencies

For every variable, its missing values will be linearly interpolated according to its closest neighbors. If they belong to a left or right NaN's tail, they will be given the value of the closest non-missing value. If they are sampled at a lower frequency than y, they be linearly interpolated to such frequency. If they are sampled at a higher frequency than y, they will be aggregated by the mean to such frequency.

2.3.2 Discarding variables dependent on the war

For each candidate variable x, we will test its independence with respect to W. Let x_0 (resp. x_1) be the values that x took when $W = 0$ (resp. $W = 1$). Let $r(x)$ be the interquartile range of a variable x. Then, x will be considered independent of W if $|\bar{x}_0 - \bar{x}_1| < 1.5\frac{r(x_0)+r(x_1)}{2}$. Graphically, this is a condition on the overlap of the two conditional histograms. Only variables that pass this test will be considered for the further steps.

2.3.3 Checking for stationarity

The target variable as well as the explicative variables as passes through an Augmented Dickey–Fuller test of stationarity. We set a confidence value $p = 0.05$. If the outcome of the test is a higher p-value, we consider the series to be stationary. Then, we create another series given by $z_t := x_t - x_{t-1}$. This differentiated series is passed again through the test. If it passes it, we keep it, otherwise, we discard it.

2.3.4 Discarding covariant variables

If several explicative variables are strongly correlated, only the one which is the most correlated with y will be kept. This step is done in order to reduce the complexity of the model while preserving the maximum information. We will consider two variables as being too correlated if their R^2 coefficient is greater or equal than 0.8.

2.3.5 Prioritizing variables and building the model

The war-independent variables will be prioritized by descending order according to its correlation with y during $\mathbb{T}_0$. These variables will be iteratively added to an optimal linear model following this order until they stop increasing its R^2 or we run out of variables.

3. Implementation and results

We implemented the methodology consistently to create a model which we will refer to by *stationary model*. However, as the reader will see, the process of making the series stationary by differentiating lead to a drastic reduce in their correlation, impacting directly the model performance. Thus, the resulting model does not attain a very good predictive power. Still, this is the correct procedure if we want to keep a statistical traceability of the errors, which is our intention.

However, and with the purpose of simply better illustrating our idea, we created a second model which softens the stationarity condition. The result is a model from which we cannot extract time-dependent statistical conclusions but which achieves a greater predictive power. This second model will be denoted by *non-stationary model* and is also explained in this report.

We start by mentioning an initial attempt to build a model based on lagged copies of the original wheat price variable, and the reason why we discarded it.

3.1 Previous attempts

In our initial modelling strategy, we estimated the effect of various economic and climatic factors on wheat prices without employing a basket of controls or considering variable selection strategies to mitigate endogeneity or omitted variable bias.

The baseline regression equation was specified as follows:

$$P_t = \alpha + \beta_1 T_t + \beta_2 T_{\text{max},t} + \beta_3 O_t + \beta_4 S_t + \beta_5 F_t + \beta_6 P_{\text{gas},t} + \beta_7 P_{\text{corn},t} + \beta_8 P_{t-1} + \gamma W_t + \epsilon_t \tag{1}$$

We also estimated a version of the model excluding the war variable W_t:

$$P_t = \alpha + \beta_1 T_t + \beta_2 T_{\text{max},t} + \beta_3 O_t + \beta_4 S_t + \beta_5 F_t + \beta_6 P_{\text{gas},t} + \beta_7 P_{\text{corn},t} + \beta_8 P_{t-1} + \epsilon_t \tag{2}$$

The difference in predicted wheat prices between these two models allows us to isolate the specific impact of the war:

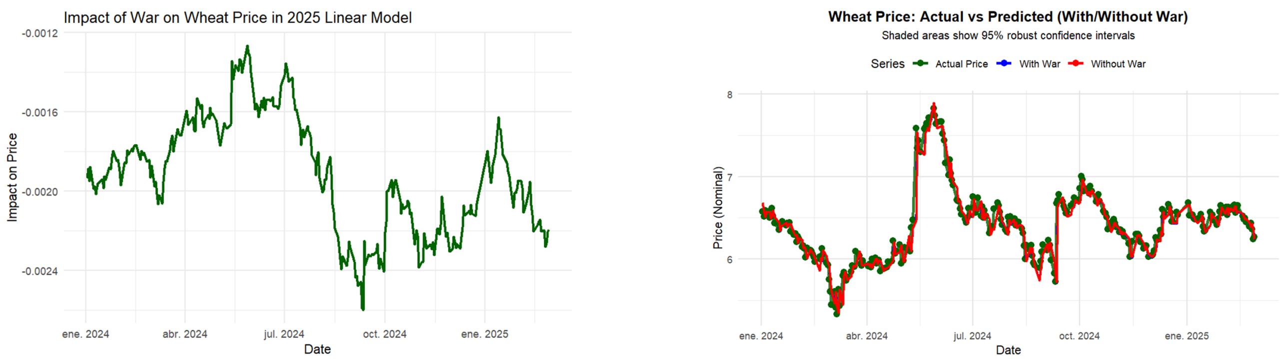
$$\Delta P_t = P_t^{\text{War}} - P_t^{\text{No War}} = \gamma W_t \tag{3}$$

This simple difference quantifies the marginal effect of the war on wheat prices, assuming all other factors are held constant. However, this approach does not account for potential confounding or spillover effects, and the exclusion of a more robust control strategy limits the reliability of these initial results.

Once the first model was done, we performed the Durbin-Watson test, the values for both models (with and without the war variable) was close to 2, indicating no significant autocorrelation in the residuals. The p-values (0.3199 and 0.3197) was for both above the 0.05 threshold, leading to a failure to reject the null hypothesis of no positive autocorrelation. This supports the assumption that the residuals are independently distributed, a key requirement for valid linear regression inference. However, tests revealed violations of other assumptions: the Breusch-Pagan test detected heteroskedasticity, and the Shapiro-Wilk test indicated non-normality in the residuals. To address these issues, we employed heteroskedasticity-consistent standard errors (HC1) for robust coefficient significance testing.

Additionally, we applied a stepwise regression approach to identify the most relevant predictors of wheat prices. Starting from a null model, we performed bidirectional selection based on AIC to achieve an optimal balance between model fit and complexity. Robust estimation was again used here to mitigate the effects of heteroskedasticity and non-normal residuals, thereby improving the reliability of the final model estimates.

Applying this procedure result on having the following:



(a) ΔP (b) Price Prediction

Figure 4: Variation and prediction of wheat prices.

Then, once the model was validated, we measured the war impact by performing an ANOVA test over the war binary variable. As can be seen from 5, the small p-value allows us to reject the null hypothesis, meaning that the war is significative.

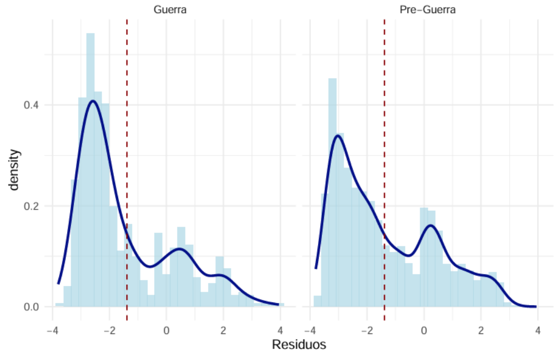
```
Analysis of Variance Table

Model 1: price_nominal_wheat ~ mean_temperature + max_temperature +
USD_Barrel_oil +
    price_nominal_soybean + fx_usd_eur + price_adjusted_gas +
    price_nominal_corn + baltic_index + P_lag
Model 2: price_nominal_wheat ~ mean_temperature + max_temperature +
USD_Barrel_oil +
    price_nominal_soybean + fx_usd_eur + price_adjusted_gas +
    price_nominal_corn + baltic_index + P_lag + war
  Res.Df    RSS Df Sum of Sq      F Pr(>F)
1   3319 52.758
2   3318 52.757  1  0.001569 0.0987 0.7534
```

Figure 5: ANOVA test for W

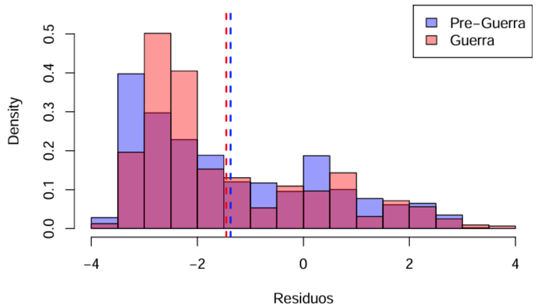
To further evaluate whether the residuals from the two models (pre-war and war-inclusive) follow the same distribution, we performed a Kolmogorov-Smirnov test, a t-test for mean equality, and Levene's test for homogeneity of variance.

The Kolmogorov-Smirnov test yielded a statistic of D = 0.1166 and a p-value = 1.205e-06, leading us to reject the null hypothesis that both sets of residuals come from the same distribution. This is visually supported by the differences observed in the density plot (Figure 6a), histogram (Figure 6b), and boxplot (Figure 7).

Despite this, the Welch Two-Sample t-test resulted in a p-value = 0.2628, suggesting that the means of



(a) Density plot of residuals from models with and without the war variable.



(b) Histogram comparison of residuals showing distribution differences.

Figure 6: Comparison of residual distributions from models with and without the war variable.

the two distributions are not significantly different. The confidence interval for the mean difference also includes zero: [-0.0613, 0.2244].

However, Levene's test for equality of variances showed significant results ($F = 14.669$, $p = 0.00013$), indicating that the variances of the residuals differ significantly between the two models.

In summary, although the residuals have similar means, they differ in distribution and variance, suggesting that the war variable introduces notable structural changes in the residual behavior.

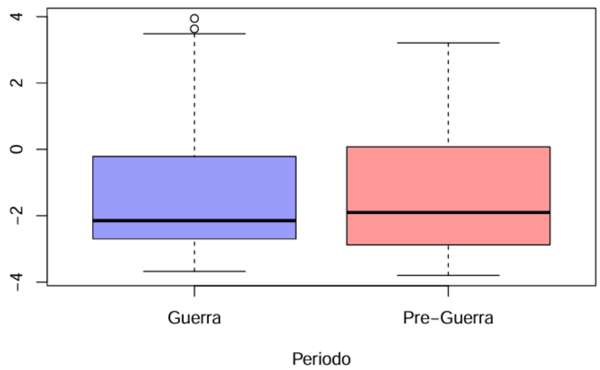


Figure 7: Boxplot of residuals by model. While medians are similar, variances differ.

In the end, this model was discarded because by including a lagged copy of the wheat price, the model really relied on it to compute the next estimate. But since the lagged wheat price also incorporates the war impact, this model was unable to isolate it.

3.2 Selecting the basket of control

The process to selecting the basket of control is common to both the stationary and non stationary model so we detail it in this previous section.

We show next the histograms of each variable conditioned to the war phase: pre-war, war beginning (first year of war) and prolonged war (after the first year of war).

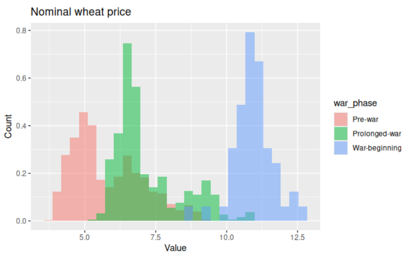


Figure 8: What price.

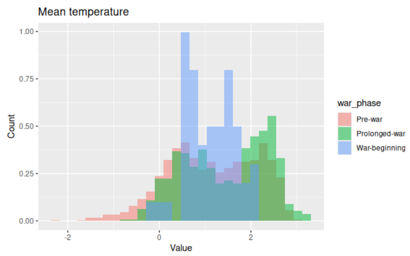


Figure 9: Mean temperature

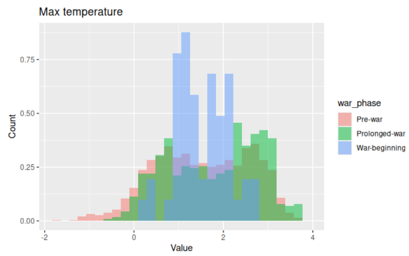


Figure 10: Maximum temperature

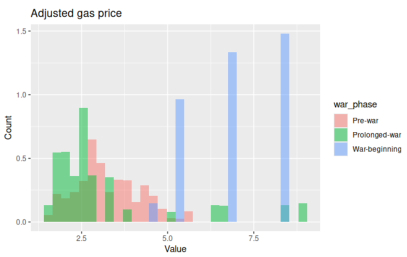


Figure 11: Adjusted gas price.

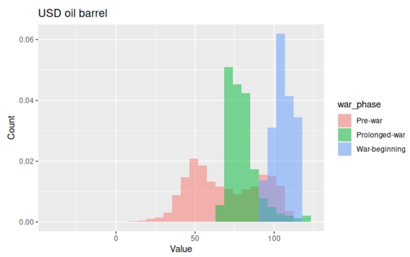


Figure 12: Oil barrel price.

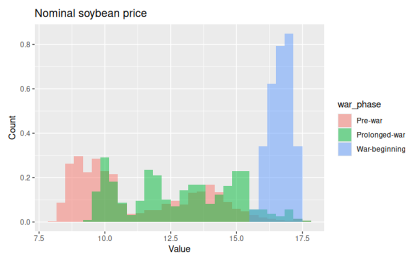


Figure 13: Nominal soybean price.

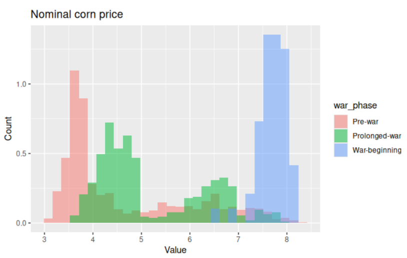


Figure 14: Nominal corn price.

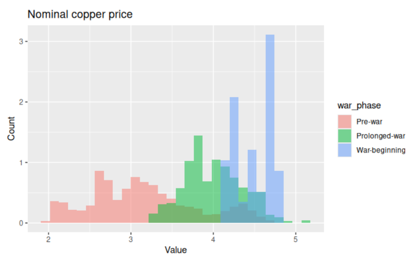


Figure 15: Nominal copper price.

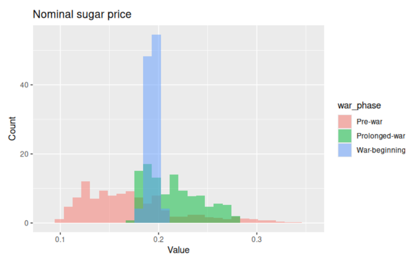


Figure 16: Nominal sugar price.

Figure 17: Nominal oats price.

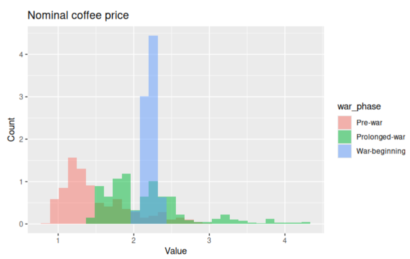


Figure 18: Nominal coffee price.

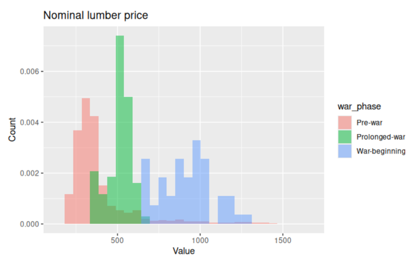


Figure 19: Nominal lumber price.

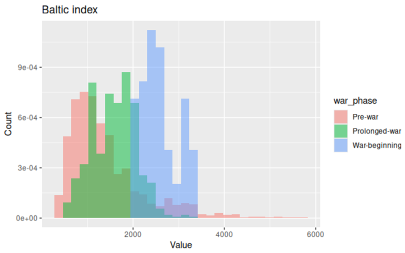


Figure 20: Baltic index.

According to the rule defined in methodology §2.3.2, the variables considered as dependent on the war are:

- Mean temperature
- Max temperature
- Oil barrel nominal price
- Sugar nominal price
- Forex EUR/USD

3.3 Stationary model

3.3.1 Stationary Checks

Before conducting time series analysis, it is essential to verify the stationarity of the data, as non-stationary series can lead to misleading inferences. Stationarity checks help ensure the statistical properties of the series remain consistent over time.

For this reason, to ensure stationarity, we applied the Augmented Dickey-Fuller (ADF) test to the available variables. The null hypothesis (H_0) is that the series has a unit root (i.e., is non-stationary), while the alternative hypothesis (H_1) is that it is stationary. We set a confidence level $p = 0.05$. The results of these tests were as follows:

Variable	**Test's p-value**
`price_nominal_wheat`	8.1×10^{-1}
`mean_temperature`	1.0×10^{-2}
`max_temperature`	1.0×10^{-2}
`price_nominal_sugar`	1.2×10^{-1}
`fx_usd_eur`	4.2×10^{-1}
`baltic_index`	5.2×10^{-2}
`USD_Barrel_oil`	5.7×10^{-1}

Table 2: Each variables p values for the stationarity test ADF.

All variables except the temperatures showed a p-value higher than our threshold. Therefore, we created new series for each one of them given by their first differences. All these new series yielded stationarity tests p-values of 10^{-2}, below our threshold so they were admitted in this new form. The temperature variables could be admitted in both forms, but we chose so consider the first-differnces series on them too for simplicity.

This lead to the following y variable and the consecutive explicative variables.

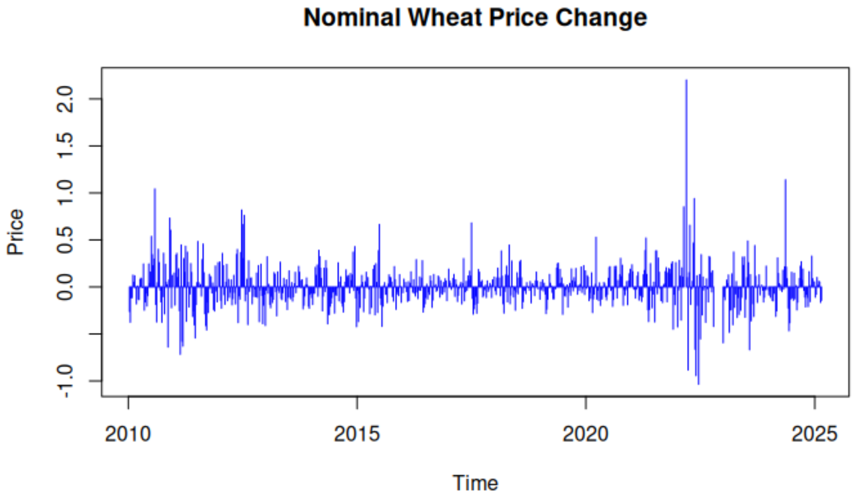


Figure 21: What price differences.

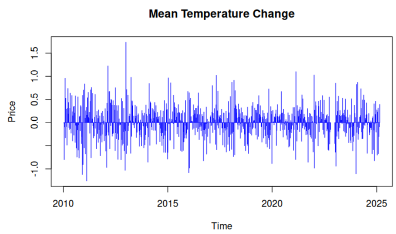


Figure 22: Mean temperature differences.

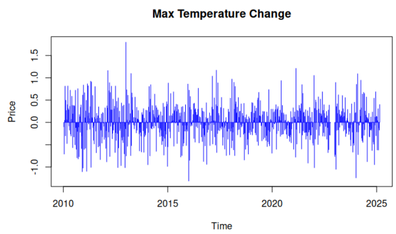


Figure 23: Max temperature differences.

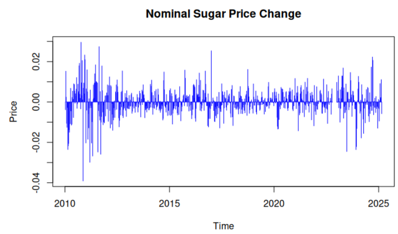


Figure 24: Sugar price differences.

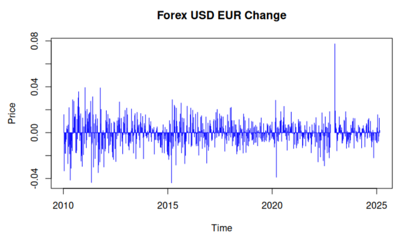


Figure 25: Forex USD EUR differences.

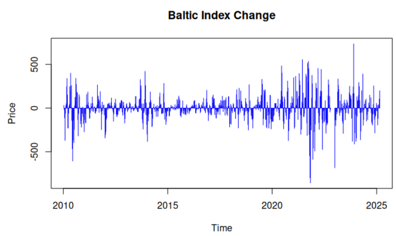


Figure 26: Baltic index differences.

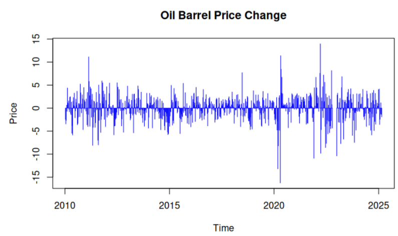


Figure 27: Oil barrel differences.

3.3.2 Selection of variables

To reduce the complexity of the model while preserving explanatory power, we identify and discard covariates that are strongly correlated with each other. When multiple explanatory variables show high pairwise correlation, we retain only the one that has the strongest correlation with the dependent variable y. This avoids multicollinearity issues and helps the model focus on the most informative predictors without introducing redundant information.

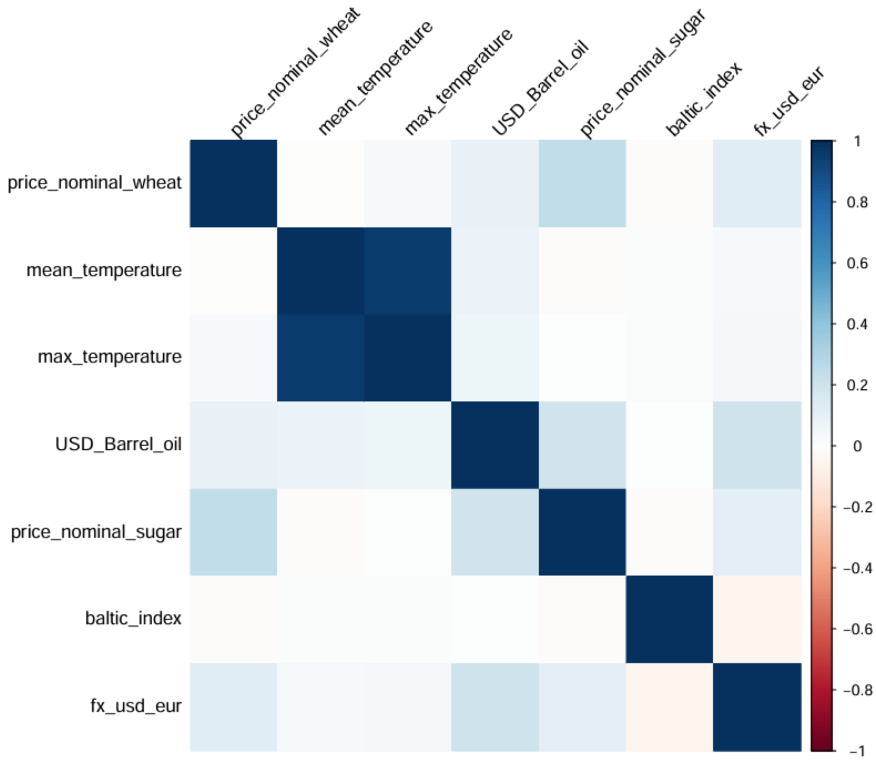


Figure 28: Correlation Matrix of Non-war affected variables

Variable	R^2
price_nominal_sugar	0.2431
fx_usd_eur	0.1239
USD_Barrel_oil	0.0906
max_temperature	0.0324
baltic_index	0.0172
mean_temperature	0.0006

Table 3: R^2 of each predictor with the dependent variable (non-stationary series).

The war-independent variables are prioritised in descending order of their correlation with the dependent variable y during time T_0. These variables are iteratively considered for inclusion in the optimal linear model. At each step, we keep the most correlated variable with y, and discard other covariates that are strongly correlated with the selected one, in order to reduce multicollinearity. This process continues until the model's R^2 no longer improves or we run out of variables.

Below is the iterative selection process:

1. **Available variables:** {`mean_temperature`, `max_temperature`, `USD_Barrel_oil`, `price_nominal_sugar`, `baltic_index`, `fx_usd_eur`}
   Selected: `USD_Barrel_oil`
   The remaining variables are not sufficiently correlated with it, so we keep them.
2. **Available variables:** {`mean_temperature`, `max_temperature`, `price_nominal_sugar`, `baltic_index`, `fx_usd_eur`}
   Selected: `fx_usd_eur`
   The remaining variables are not sufficiently correlated with it, so we keep them.
3. **Available variables:** {`mean_temperature`, `max_temperature`, `price_nominal_sugar`, `baltic_index`}
   Selected: `price_nominal_sugar`
   The remaining variables are not sufficiently correlated with it, so we keep them.
4. **Available variables:** {`mean_temperature`, `max_temperature`, `baltic_index`}
   Selected: `baltic_index`
   The remaining variables are not sufficiently correlated with it, so we keep them.
5. **Available variables:** {`mean_temperature`, `max_temperature`}
   Selected: `mean_temperature`
   `max_temperature` is highly correlated with it and is therefore discarded.
6. **Available variables:** {}
   End of selection process.

Then, the final selected variables are: `USD_Barrel_oil`, `fx_usd_eur`, `price_nominal_sugar`, `baltic_index`, and `mean_temperature`.

3.3.3 Building the model

Once the variables are selected, a linear regression model is applied. The model yields a relatively low[1] R^2 value (0.06933), indicating that only about 6.9% of the variation in wheat prices is explained by the included predictors. Despite this, the model is statistically significant overall (F-statistic = 9.222, p-value ¡ 0.001), suggesting that at least one of the predictors has a meaningful linear relationship with the dependent variable.

Alternatively, the low R^2 could be due to the process of making the series stationary. What we observe is a general pattern: when soybean prices are high, wheat prices tend to be high as well. However, this

[1] Due to data leakage related to the war, the model's predictive power is limited, as only a few reliable variables remain available for analysis.

does not imply that if soybean prices increase today, wheat prices will rise on the same day. The time it takes for such price changes to propagate may weaken correlations in differenced (stationary) series, even though the relationship persists in the original price series. We have observed that as temporal frequency decreases (monthly-weekly-daily difference), correlations tend to increase. This is because if soybean prices rise today and wheat prices rise ten days later, the relationship becomes more visible when examining aggregate monthly changes. However, lowering the frequency of the data comes at the cost of losing many observations, which is why we only considered doing so at weekly or higher levels

Looking at individual coefficients, `price_nominal_sugar` shows a highly significant positive relationship with wheat prices ($p < 0.0001$), while `fx_usd_eur` is significant at the 5% level. Other variables, such as oil prices, the Baltic index, and mean temperature, do not show statistically significant effects ($p > 0.1$).

Residuals appear centered around zero with a relatively low residual standard error (0.2013), suggesting acceptable dispersion.

```
##
## Call:
## lm(formula = price_nominal_wheat ~ USD_Barrel_oil + fx_usd_eur +
##     price_nominal_sugar + baltic_index + mean_temperature, data = df0_diff)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -0.6933 -0.1266 -0.0185  0.1167  1.0255
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)          6.800e-03  8.062e-03   0.844   0.3993
## USD_Barrel_oil       2.131e-03  3.192e-03   0.668   0.5047
## fx_usd_eur           1.610e+00  6.922e-01   2.325   0.0204 *
## price_nominal_sugar  6.772e+00  1.183e+00   5.726  1.6e-08 ***
## baltic_index        -1.305e-05  5.382e-05  -0.242   0.8085
## mean_temperature    -1.267e-03  2.136e-02  -0.059   0.9527
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2013 on 619 degrees of freedom
## Multiple R-squared:  0.06933,    Adjusted R-squared:  0.06181
## F-statistic: 9.222 on 5 and 619 DF,  p-value: 1.765e-08
```

Figure 29: R Output of Stationary Linear Model

The plot of date versus price differences highlights a noticeable divergence between predicted and observed values during the war period. Specifically, the model fails to capture several sharp fluctuations in wheat prices, leading to substantial residuals in this timeframe. This discrepancy can be partially attributed to the loss of relevant information when differencing the series to achieve stationarity. As previously discussed, important relationships—such as the lagged influence of soybean prices on wheat—may no longer be detectable in the differenced data. These delayed effects, which might manifest over several days or weeks, are smoothed out or lost when focusing solely on short-term variations. Moreover, the war likely introduced sudden shocks and structural changes that are difficult for a linear model to anticipate, especially when using only a limited set of stationary predictors.

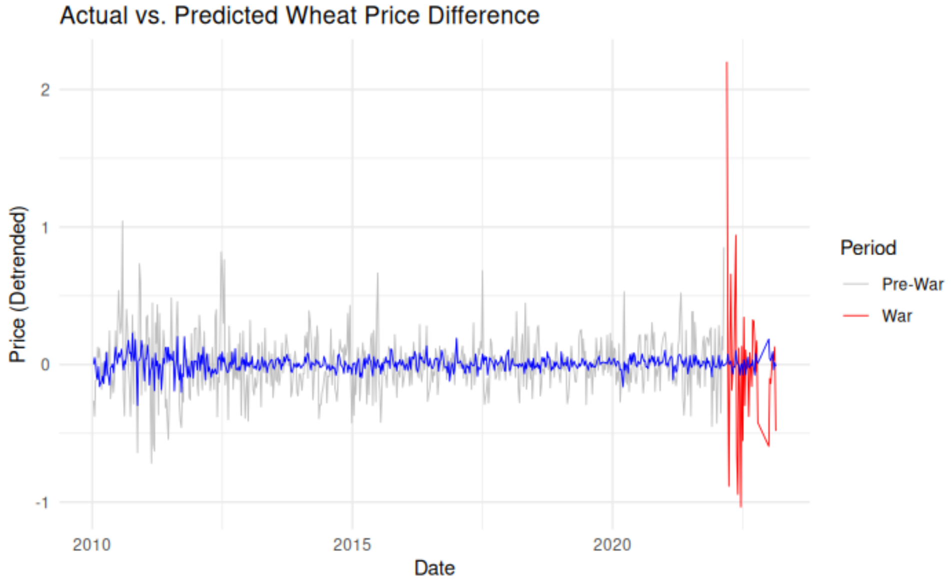


Figure 30: Stationary model predictions over $\mathbb{T}$.

With the forecasted difference in the wheat price, we can reconstruct the total price predictions of our model by cumulative sum. We observe that the loss of accuracy predicting weekly wheat price changes is translated to a loss of accuracy predicting the full price. See figure 31.

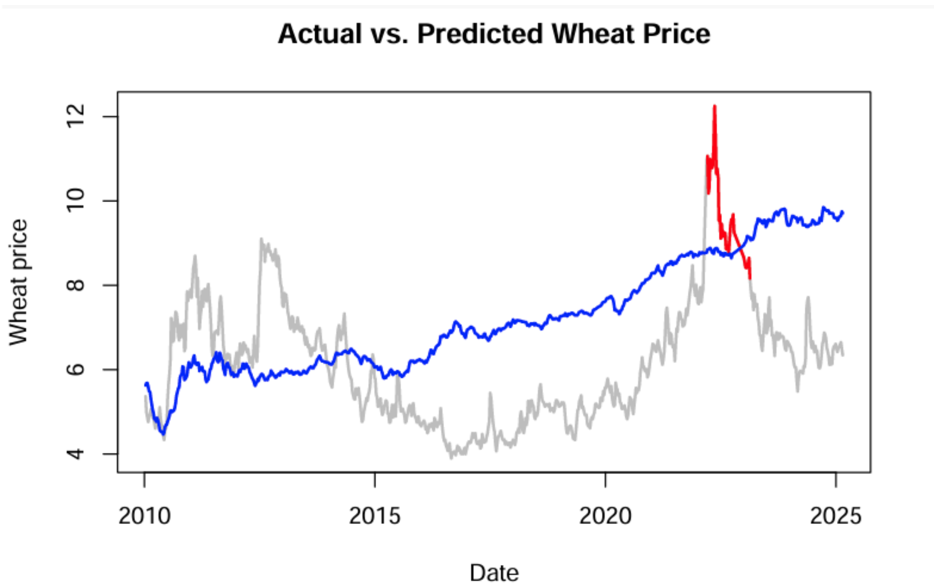


Figure 31: Stationary model predictions over $\mathbb{T}$.

3.3.4 Estimating the ε distribution

Following the methodology (§2.1), we now proceed to estimate the distribution of the model errors ε during $\mathbb{T}_0$. We assume that $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ and compute σ so that it maximizes the likelihood of the data. The result is $\sigma = 0.2003$.

Recall that according to our methodology, the error during $\mathbb{T}_0$ equals ε, while the error during $\mathbb{T}_1$ equals $\varepsilon + W$. Figure 32 shows the distribution of the errors during $\mathbb{T}_0$ and $\mathbb{T}_1$. A curve in blue shows the density curve of a $\mathcal{N}(0, \sigma^2)$. The distribution of ε in $\mathbb{T}_0$ fits well enough this normal distribution. However, during $\mathbb{T}_1$, it is more spreaded, indicating that the model failed at $\mathbb{T}_1$ more than it did at $\mathbb{T}_0$ therefore W was far away from null.

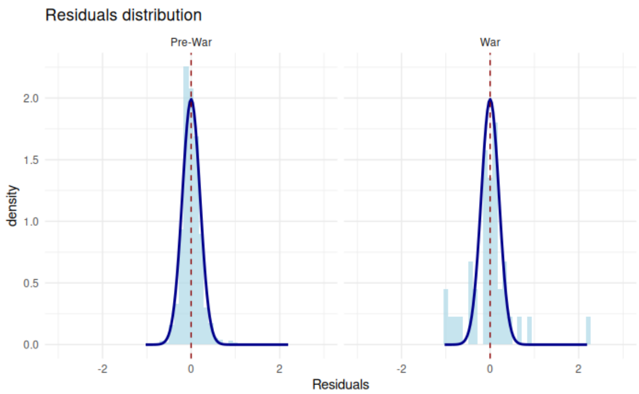


Figure 32: Residuals of the stationary model during $\mathbb{T}_0$ and $\mathbb{T}_1$.

3.3.5 Estimating W

We now proceed to compute a stochastical lower bound of W during $\mathbb{T}_1$ as explained in the methodology §2.1. Setting a confidence value of $p = 0.05$, we want to find ε_0 such that $P(|\mathcal{N}(0, \sigma^2)| < \varepsilon_0) > p$. We obtain $\varepsilon_0 = 0.3296$. With this, for every time $t \in \mathbb{T}_1$, $y_t - \hat{y}_t - \varepsilon_0 < W$ at confidence p, so it is the lower bound we are looking for.

With this setting, the lower bounds of W_t are shown in

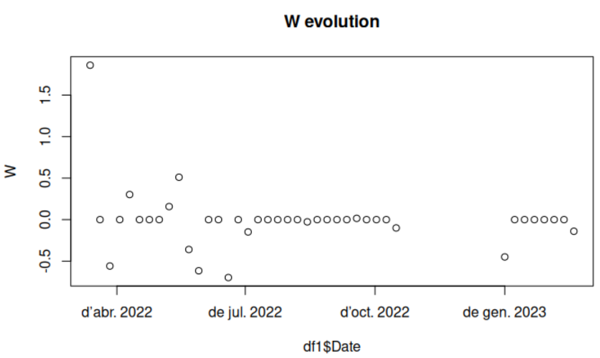


Figure 33: Observed W lower bound across $t \in \mathbb{T}_1$.

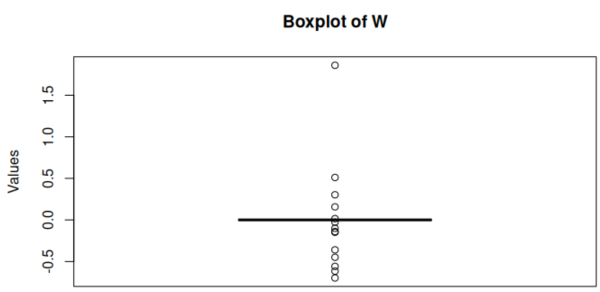


Figure 34: Observed W lower bounds distribution.

We observe that the errors, as seen in figure 35, lie almost entirely within $[-\varepsilon_0, \varepsilon_0]$. During the war, there are a few instances when the errors really exceed this stochastic threshold, meaning that then, with confidence of $1 - p = 0.95$, W was different than 0.

Finally, the war impact is measured according to the methodology, since we are working with a differentiated time series as

$$\sum_{t \in \mathbb{T}_1} W_t = -0.2532.$$

From this, one may erroneously extract the conclusion that the overall shifts that the war is due to have produced at confidence p were of at least -0.2532\$/Bushel. However, if one examines closely our model, one sees that its R^2 equals 0.0618, which means it has no predictive power. Not only does it constantly fail to predict the next price increase, but it even overestimates the price increase in the end of the first year of war. Therefore, we would be attributing to the war any price variation that exceeds the likely values it had previously to war. Our methodology is only consistent as long as the model actually captures the impact that the other variables had on y. Here, it is not capturing anything at all. Therefore, the only conclusion we can extract from this is that the data we considered for this project is not able to linearly explain the weekly variations of the wheat price. And next, no proposition can be extracted following this initial failure.

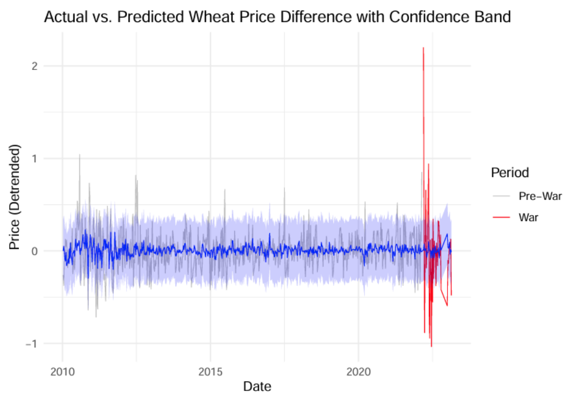


Figure 35: Stationary model's predictions with a confidence of $1 - p = 0.95$.

3.4 Non stationary model

3.4.1 Selection of variables

At this point, variables have already been discarded based on their dependence with respect to the war. Now, we compute their correlation matrix. Based on this, we will iteratively select the most wheat-correlated one (x) and discard the variables which are too correlated with x. This is performed complying with methodology §2.3.4 and §2.3.5.

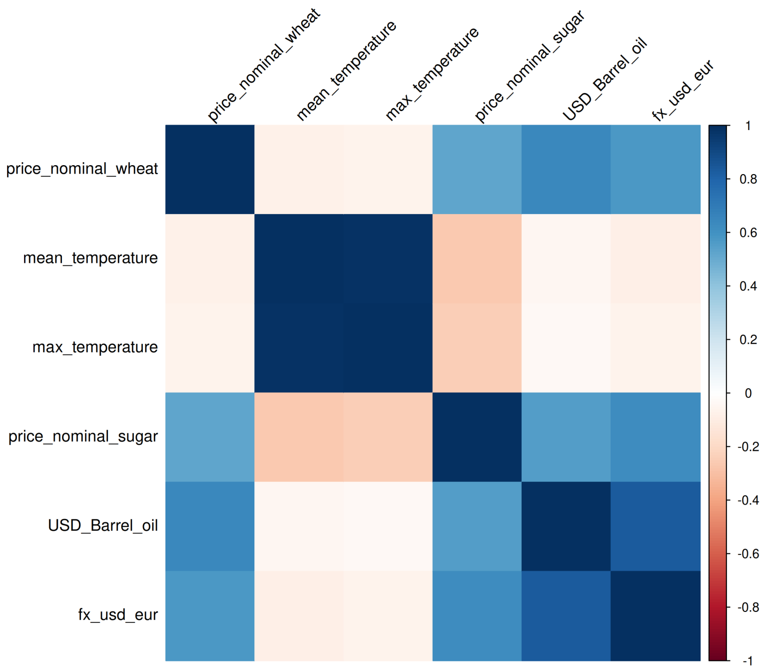


Figure 36: Correlation matrix of the basket of control variables without differentiating, together with the wheat price variable.

The selection process for the model's variables was based on their correlation with the target variable y and the redundancy among themselves.

Starting with the initial pool of variables—`mean_temperature`, `max_temperature`, `price_nominal_sugar`, `USD_Barrel_oil`, and `fx_usd_eur`—the variable most correlated with y was found to be `USD_Barrel_oil`. Although `fx_usd_eur` showed a relatively high correlation with `USD_Barrel_oil` (correlation coefficient 0.8387, $R^2 = 0.7032$), this value was below the pre-established threshold of 0.8, so both variables were retained.

Removing `USD_Barrel_oil`, the next variable with the strongest correlation to y was `fx_usd_eur`, with the remaining variables exhibiting low correlation with it, so all were kept.

Continuing the process, `price_nominal_sugar` emerged as the variable most correlated with y, and similarly, the other variables were not highly correlated with it, so none were excluded.

Finally, when considering mean_temperature and max_temperature, it was observed that these two were very strongly correlated ($r = 0.9888$, $R^2 = 0.9777$), exceeding the redundancy threshold. Consequently, max_temperature was removed from the model.

Thus, the final set of selected variables included USD_Barrel_oil, fx_usd_eur, price_nominal_sugar, and mean_temperature.

We show in table 4 the R^2 coefficient of each selected variable with respect to y.

Variable	R^2
USD_Barrel_oil	0.6424
fx_usd_eur	0.5702
price_nominal_sugar	0.5242
mean_temperature	0.0747

Table 4: Variables and their R^2 with y.

3.4.2 Building the model

Before creating the model, all variables got their linear trend removed. Then, a linear model is fit on $\mathbb{T}_0$ and then used to predict y for all $\mathbb{T}$. The results obtained are shown in figure 37. The models coefficients are shown in table

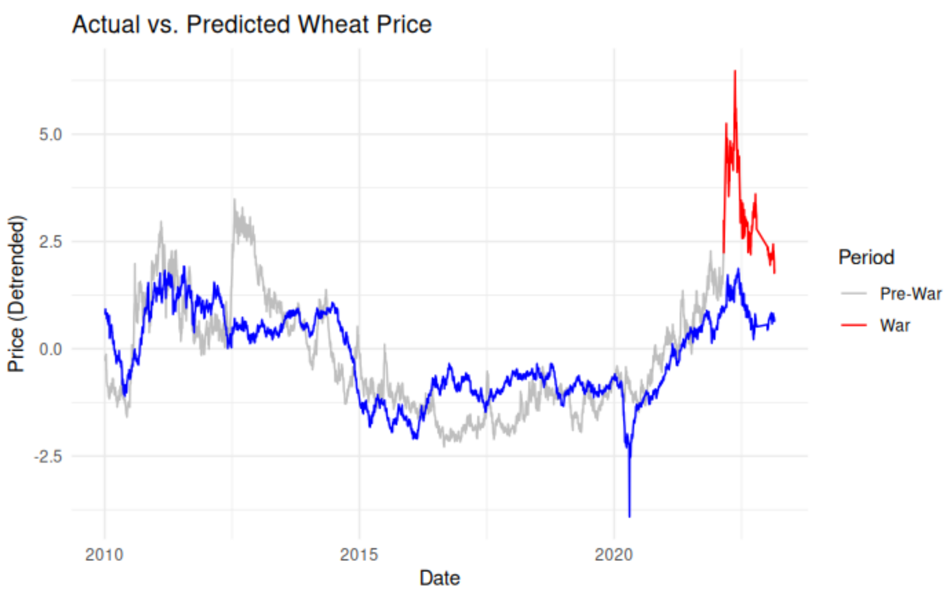


Figure 37: Non stationary model predictions over $\mathbb{T}$.

As we can see in figure 37, the model underestimates the wheat price during the war. Particularly high is the gap during the beginning of the war. This model attains an R^2 with y of 0.5446.

```
## Call:
## lm(formula = price_nominal_wheat ~ USD_Barrel_oil + fx_usd_eur +
##     price_nominal_sugar + mean_temperature, data = df0_detrended)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.8713 -0.6647 -0.0276  0.5275  3.1785
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)         -0.033780   0.016793  -2.012   0.0444 *
## USD_Barrel_oil       0.029831   0.001179  25.307  < 2e-16 ***
## fx_usd_eur           1.819926   0.289741   6.281 3.86e-10 ***
## price_nominal_sugar  9.117945   0.408070  22.344  < 2e-16 ***
## mean_temperature     0.031055   0.017189   1.807   0.0709 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.868 on 2928 degrees of freedom
## Multiple R-squared:  0.5452, Adjusted R-squared:  0.5446
## F-statistic: 877.5 on 4 and 2928 DF,  p-value: < 2.2e-16
```

Figure 38: R Output of Non Stationary Linear Model

3.4.3 Estimating the ε distribution

Following the methodology (§2.1), we now proceed to estimate the distribution of the model errors ε during $\mathbb{T}_0$. We assume that $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ and compute σ so that it maximizes the likelihood of the data. The result is $\sigma = 0.9810$.

Recall that according to our methodology, the error during $\mathbb{T}_0$ equals ε, while the error during $\mathbb{T}_1$ equals $\varepsilon + W$. Figure 39 shows the distribution of the errors during $\mathbb{T}_0$ and $\mathbb{T}_1$. A curve in blue shows the density curve of a $\mathcal{N}(0, \sigma^2)$. The distribution of ε in $\mathbb{T}_0$ fits well enough this normal distribution. However, during $\mathbb{T}_1$, it is shifted to the left, indicating that W was far away from null.

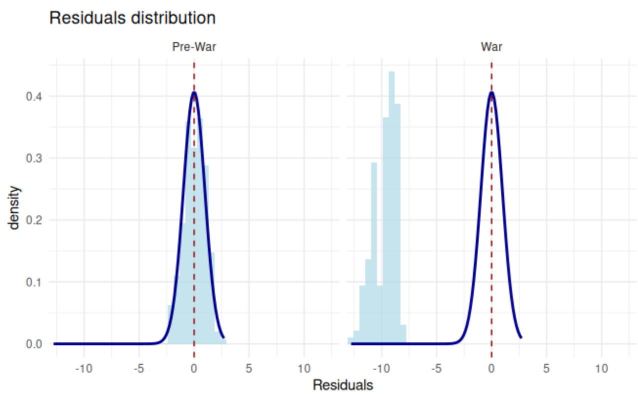


Figure 39: Residuals of the non stationary model during $\mathbb{T}_0$ and $\mathbb{T}_1$.

3.4.4 Estimating W

We now proceed to compute a stochastical lower bound of W during $\mathbb{T}_1$ as explained in the methodology §2.1. Setting a confidence value of $p = 0.05$, we want to find ε_0 such that $P(|\mathcal{N}(0, \sigma^2)| < \varepsilon_0) > p$. We obtain $\varepsilon_0 = 1.6136$. With this, for every time $t \in \mathbb{T}_1$, $y_t - \hat{y}_t - \varepsilon_0 < W$ at confidence p, so it is the lower bound we are looking for.

With this setting, the lower bound of W_t are shown in figures 40 and 41.

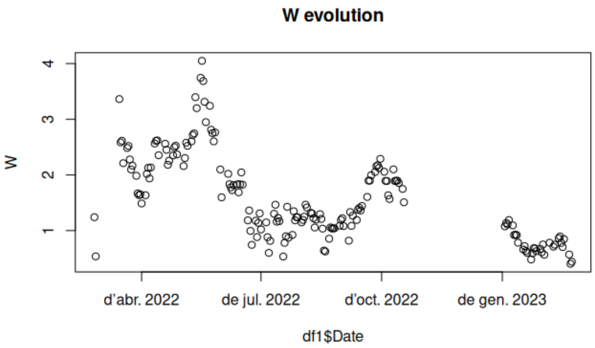


Figure 40: Observed W lower bound across $t \in \mathbb{T}_1$.

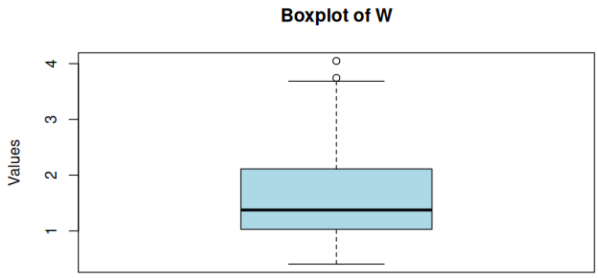


Figure 41: Observed W lower bounds distribution.

We can see that in almost all instances we can find a positive lower bound for W, indicating that the gap at such time was so high that it is unlikely that it was due to the original variability. The gaps are positive since this means that the real price was higher than what would have been under normal circumstances.

Finally, the war impact is measured according to the methodology as

$$\frac{1}{\mathbb{T}_1} \sum_{t \in \mathbb{T}_1} W_t = 1.5842.$$

We conclude that the average impact of the Russian invasion of Ukraine was at least of 1.5842\$/Bushel.

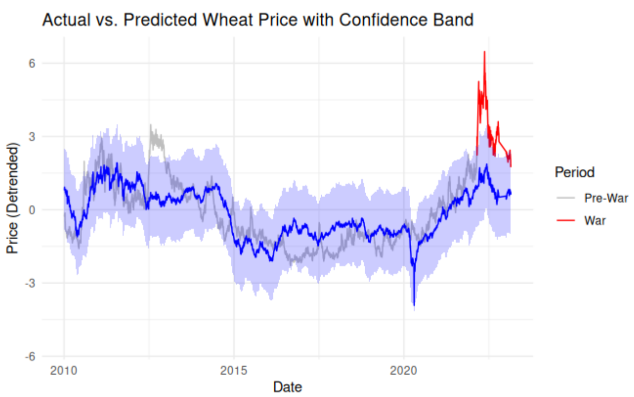


Figure 42: Non stationarity model's predictions with a confidence of $1 - p = 0.95$.

It is very interesting to notice that, at this confidence level, the model only fails during 2013, when the Euromaidan conflict occured. So the only two moments in the recent years where the wheat price series stochastical law changed coincide with armed conflicts between Russia and Ukraine.

4. Conclusions

This study set out to quantify the impact of the 2022 Russian invasion of Ukraine on global wheat prices through the construction of a predictive linear model. Despite observing clear price increases in the post-invasion period, our findings show that the predictive power of linear models is highly sensitive to the transformation of the input data and the underlying assumptions required for causal inference.

When differencing the time series to ensure stationarity, we are able to apply standard linear modeling techniques with statistically valid inference. However, the resulting model displays very limited explanatory power. The low R^2 value (6.9%) suggests that only a small portion of the variation in wheat prices can be explained by the included predictors in this transformed space. Moreover, during the war period, predicted values differ markedly from observed prices. This poor performance is likely due to the loss of long-term dependencies and delayed effects between variables, such as the lagged transmission from soybean prices to wheat prices, which are obscured when daily differences are used. As a result, the model is incapable of capturing the structural shifts induced by the war, and we conclude that, under these conditions, it is not possible to construct a linear model with sufficient predictive power to isolate or quantify the war's impact.

In contrast, models constructed from the original, non-differenced (non-stationary) data provide better predictive fit. However, these models lack the temporal resolution needed for causal analysis. By discarding stationarity, we also forfeit the ability to make rigorous inferences about localized time periods such as the war window. In these models, relationships are interpreted globally across the entire time span (2000–2024), meaning that any attribution of price behavior to the war is speculative, especially given that error distributions are not guaranteed to be stable across time.

Ultimately, while deviations between predicted and actual prices during the war period suggest an abnormal external influence—presumably from the invasion itself—our methodology is only able to offer lower-bound estimates of this effect under high-confidence assumptions. The counterfactual analysis, constrained by the need for uncontaminated explanatory variables and the limitations of linear models, reveals the inherent difficulty of isolating the war's impact in a highly interdependent global market.

Thus, we conclude that:

1. Stationary models cannot reliably quantify the war's impact due to the loss of essential dynamics and poor predictive performance given the variables we considered.
2. Non-stationary models, while better fitting, lack inferential reliability and temporal precision for causal claims.
3. Causal inference in this context is deeply constrained by variable selection, model structure, and the systemic nature of global commodity interactions.

Therefore, despite strong economic intuition and observed price surges, the specific effect of the war on wheat prices cannot be precisely measured using our approach. Any quantification must be interpreted as a conservative estimate, acknowledging the limitations of both the data and the modeling framework.

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